

Total Cholesterol and Cancer Risk in a Large Prospective Study in Korea

Purpose To further clarify the relationship between total cholesterol and cancer, which remains unclear. **Methods** We prospectively examined the association between total cholesterol and site-specific and all-cancer incidence among 1,189,719 Korean adults enrolled in the National Health Insurance Corporation who underwent a standardized biennial medical examination in 1992 to 1995 and were observed for 14 years until cancer diagnosis or death. **Results** Over follow-up, 53,944 men and 24,475 women were diagnosed with a primary cancer. Compared with levels less than 160 mg/dL, high total cholesterol (≥ 240 mg/dL) was positively associated with prostate cancer (hazard ratio [HR], 1.24; 95% CI, 1.07 to 1.44; P trend = .001) and colon cancer (HR, 1.12; 95% CI, 1.00 to 1.25; P trend = .05) in men and breast cancer in women (HR, 1.17; 95% CI, 1.03 to 1.33; P trend = .03). Higher total cholesterol was associated with a lower incidence of liver cancer (men: HR, 0.42; 95% CI, 0.38 to 0.45; P trend < .001; women: HR, 0.32; 95% CI, 0.27 to 0.39; P trend < .001), stomach cancer (men: HR, 0.87; 95% CI, 0.82 to 0.93; P trend $\leq$.001; women: HR, 0.86; 95% CI, 0.77 to 0.97; P trend = .06), and, in men, lung cancer (HR, 0.89; 95% CI, 0.82 to 0.96; P trend < .001). Results for liver cancer were slightly attenuated after additional adjustment for liver enzyme levels and hepatitis B surface antigen status (men: HR, 0.60; P trend < .001; women: HR, 0.46; P trend = .003) and exclusion of the first 10 years of follow-up (men: HR, 0.59; P trend < .001; women: HR, 0.44; P trend < .001). Total cholesterol was inversely associated with all-cancer incidence in both men (HR, 0.84; 95% CI, 0.81 to 0.86; P trend < .001) and women (HR, 0.91; 95% CI, 0.87 to 0.95; P trend < .001), but these associations were attenuated after excluding incident liver cancers (men: HR, 0.95; P trend < .001; women: HR, 0.98; P trend = .32). **Conclusion** In this large prospective study, we found that total cholesterol was associated with the risk of several different cancers, although these relationships differed markedly by cancer site.